Saturday Test — No. 4

Take your time. Show your work where it helps.

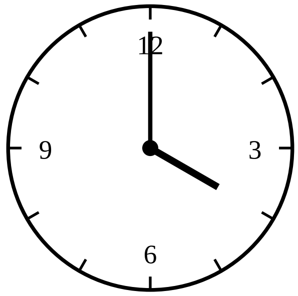
Name: ____________________ *Date:* __________ *Time: 15 min*

Numbers and Shapes

1. Suppose **A** and **B** stand for two unknown numbers. If **A + A + A = 18** and **A + B = 11**, what is **B**?

2. What is the largest two-digit number whose two digits multiply together to give 12?

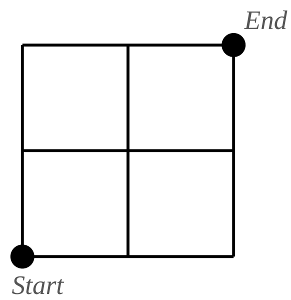
3. At exactly 4 o’clock, what is the angle between the hour hand and the minute hand?



4. A red light blinks every 6 seconds. A blue light blinks every 8 seconds. They blink together at time zero. After how many seconds will they next blink at exactly the same moment?

Counting Carefully

5. On the grid below, walk from *Start* to *End* by moving only **right** or **up** along the lines. How many different paths are there?



6. In a class of 20 students, 12 like apples, 9 like bananas, and 4 like both. How many students like neither apples nor bananas?

Reasoning

7. On an island, every person is either a *knight* (who always tells the truth) or a *knave* (who always lies). You meet two islanders, **Pat** and **Quinn**.

Pat says: “We are both knights.”
Quinn says: “Pat is lying.”

Who is the knight and who is the knave? Explain how you know.

8. You have a **5-litre jug** and a **3-litre jug**, both empty, and a tap with all the water you want. Neither jug has any markings on it. How can you measure out exactly **4 litres** of water? Describe the steps.

Answer Key — for parents

Numbers and Shapes

1. A + A + A = 18 means A = 6. Then A + B = 11, so B = 11 – 6 = **5**.

2. Digit pairs that multiply to 12: 2×6, 3×4 (and 6×2, 4×3). The two-digit numbers are 26, 34, 43, and 62. The largest is **62**.

3. A full turn is 360°, so each of the 12 hour-marks is 360° ÷ 12 = 30°. At 4 o'clock the hour hand is at the 4 and the minute hand is at the 12, four hour-marks apart: 4 × 30° = **120°**.

4. The next moment they blink together is the smallest number that is a multiple of both 6 and 8 (the LCM). Multiples of 8: 8, 16, **24**, … and 24 = 6 × 4. Answer: **24 seconds**.

Counting Carefully

5. Each path uses 2 right-moves and 2 up-moves in some order — choose which 2 of the 4 steps are rights. C(4,2) = **6 paths**. (RRUU, RURU, RUUR, URRU, URUR, UURR.)

6. Students who like apples or bananas (or both): 12 + 9 – 4 = 17 (we subtract the 4 to avoid double-counting). Students who like neither: 20 – 17 = **3**.

Reasoning

7. Suppose Pat were a knight. Then "we are both knights" would be true, making Quinn a knight too — but Quinn says Pat is lying, which a knight would never say about a fellow knight. Contradiction, so Pat must be a knave. Then Pat's claim is false, which is consistent. Quinn's claim "Pat is lying" is true, so Quinn is a knight. **Pat is the knave; Quinn is the knight.**

8. One solution: (1) Fill the 5-L jug from the tap. (2) Pour from the 5-L into the 3-L until the 3-L is full — the 5-L now holds 2 L. (3) Empty the 3-L jug. (4) Pour the 2 L from the 5-L into the 3-L (now the 3-L holds 2 L). (5) Refill the 5-L from the tap. (6) Pour from the 5-L into the 3-L until the 3-L is full — only 1 L fits, leaving exactly **4 L in the 5-L jug**.